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Literature Review and Research Redesign for the Human-Like Adaptive Visual Attention Project

Where this line of work already stands

Your project sits at the intersection of three mature literatures and one younger one. The mature literatures are: visual saliency and fixation prediction; model–brain alignment in vision; and representational comparison methods such as encoding models and RSA. The younger literature is the attempt to connect **human-like attention**, **adaptive/foveated computation**, and **brain alignment** in one framework. That distinction matters, because it means your project is not asking a question the field has fully settled, yet every major subproblem has already been studied in depth and with recognizable methodological conventions. [1]

Historically, the "saliency" idea entered computational neuroscience through accounts such as Li's V1 saliency-map proposal, which treated saliency as a mechanism for bottom-up attentional priority. In parallel, the neuroimaging community developed encoding and decoding models for fMRI, and Kriegeskorte's RSA framework supplied a general method for comparing representations across brains, models, and measurement modalities. The modern neuro-AI line then accelerated when Yamins and colleagues showed that performance-optimized hierarchical models could predict higher visual cortex, and later argued more broadly for goal-driven deep learning as a strategy for understanding sensory cortex. [2]

From there, the field became benchmark-centered. Brain-Score formalized the idea that a good candidate model of vision should explain both neural and behavioral data on a common evaluation platform, and NSD made large-scale fMRI modeling over natural scenes feasible by providing whole-brain 7T data over thousands of images and multiple sessions per subject. Algonauts translated that infrastructure into an open challenge format, which strongly shaped current conventions in brain-alignment work: standardized stimuli, public leaderboards, held-out prediction, and direct model comparison. [3]

That history already suggests an important framing point for your project. The field no longer treats "brain-likeness" or "human-like attention" as a single scalar property. It increasingly treats them as partially separable dimensions: voxelwise predictivity, representational geometry, behavioral similarity, task performance, robustness, and sometimes efficiency. This is exactly the conceptual move your project should make explicit. [4]

What the saliency and fixation literature has already established

The saliency literature is much more advanced than a simple comparison of explanation heatmaps against human fixation maps. DeepGaze I and II helped establish that transfer learning from object-recognition backbones could substantially improve fixation prediction, and DeepGaze IIE showed that calibration and out-of-domain generalization were major issues even when in-domain scores were strong. DeepGaze III then moved from static saliency maps toward **scanpath prediction**, explicitly conditioning future fixations on fixation history rather than predicting only a one-shot importance map. The 2023 Annual Review by Kümmerer and Bethge makes clear that the field now distinguishes saliency-map modeling from scanpath modeling and treats probabilistic evaluation as the most principled common language across both settings. [5]

Current SOTA also looks different from what many older saliency papers would lead one to expect. The recent DeepGaze MSDB work argues that while fixation models are approaching gold-standard performance on individual benchmarks, cross-dataset generalization is still fragile because dataset bias is large. In that paper, transfer to unseen saliency datasets drops by roughly 40%, and close to 60% of the inter-dataset gap is attributed to dataset-specific biases rather than some single universal fixation predictor. The model addresses this by separating mostly dataset-agnostic features from a small number of dataset-specific bias parameters and sets the benchmark high across MIT300, CAT2000, and COCO-Freeview. This finding is directly relevant to your project because it means that differences across SALICON, CAT2000, and COCO-Search-like data should be interpreted as substantive, not as noise. [6]

A second major lesson from this literature is that **free-viewing**, **task-driven search**, and **scanpath generation** are not interchangeable experimental regimes. The fixation-review literature explicitly separates them, and the newer scanpath literature increasingly models temporal order, saccadic momentum, and prior fixation history. That matters for your project because a model that looks "human-like" under NSS on free-viewing saliency maps is not necessarily human-like under search or sequential gaze metrics. DeepGaze III and related work are strong evidence that if your project ultimately wants to speak about "adaptive visual attention," scanpath-level or sequential analyses will eventually matter, not only static heatmaps. [7]

There is also a clear methodological convention in this field that your project should adopt. Serious saliency papers nearly always report strong baselines such as center bias, compare against human upper bounds or leave-one-subject-out gold standards, and avoid treating raw saliency metrics as architecture-agnostic absolutes across datasets. The public MIT/Tübingen benchmark encodes this culture explicitly. Your current behavioral benchmark is moving in the right direction when it includes center-bias controls and DeepGaze rows, but the literature makes clear that these controls are not optional—they are the minimum standard. [8]

What the human–machine attention literature says about relevance to your project

A large part of the attention-alignment literature is really about a distinction your project must keep clear: **human fixation patterns**, **post-hoc explanation maps**, and **internal transformer routing** are related objects, but they are not the same object. Grad-CAM was proposed as a class-localization explanation for CNNs; RISE was proposed as a black-box importance estimator using randomized masking; attention rollout/flow was developed because raw transformer attention weights are unreliable explanation probes; Chefer et al. proposed relevance propagation beyond attention visualization; and AttnLRP pushed transformer attribution further toward faithfulness and efficiency. In other words, the mainstream interpretability literature already warns against conflating "attention" with "what the model truly uses." Your proposal's decomposition of machine saliency into different families is therefore methodologically correct and should be preserved, then expanded. [9]

That caution becomes even stronger in vision transformers. Mehrani and Tsotsos argue that ViT self-attention behaves more like perceptual grouping than human visual attention, and Abnar and Zuidema show that information gets mixed through layers in ways that make raw attention weights poor explanations. This does not mean ViTs cannot become human-aligned. It means that if you want to make claims about human-like attention, you need to compare multiple attribution families and distinguish **internal routing** from **behavioral saliency alignment**. A paper that treats ViT attention maps themselves as evidence of human-like attention would be methodologically vulnerable in the current literature. [10]

The literature comparing human and machine visual strategies also cuts both for and against your current direction. On the supportive side, Fel et al. studied 84 ImageNet DNNs and found that higher classification accuracy systematically traded off against alignment with human visual strategies, then showed that a harmonization procedure could increase both alignment and accuracy. Müller et al. showed that human–CNN similarity depends heavily on the **task used to elicit human attention** and on image type; for some image classes, manual relevance annotations align better with CNN evidence than spontaneous eye movements do. Together, these results support your intuition that fixation data and machine saliency are scientifically meaningful to compare, yet they also warn that "alignment" depends strongly on the human data collection paradigm. [11]

On the more encouraging side for your project, there is now credible evidence that human-like attention can **emerge** or be **induced** in machine vision systems. Yamamoto et al. report that self-supervised ViTs trained with DINO-like objectives develop distinct attention-head clusters, with some late-layer heads focusing on regions that resemble human gaze patterns in video. Kockwelp et al. show that human gaze can be used for token masking in ViTs to improve performance while pushing processing toward more human-relevant regions. Koorathota et al. use a fixation-attention intersection loss to improve ViT decision-making under uncertainty in driving, aligning attention with human gaze. Schwinn et al.'s NeVA work shows that a biologically inspired foveated constraint can produce more human-like scanpaths even without direct scanpath supervision. These papers collectively support the **direction** of your project: human-like overt attention is not a dead end, and adaptive or selective computation is a plausible route into the problem. [12]

At the same time, the strongest adaptive-attention papers in 2025–2026 are still mostly **emerging**, and many are either workshops, preprints, or under review. Examples include hard-attention models such as EVA, saccade-inspired ViT pipelines, and foveated brain-encoding proposals such as MindAttention. These works are useful because they show where the field is heading: beyond static heatmaps and toward active, sequential, resource-constrained vision. Yet they are not stable enough to be treated as settled foundations. For your project, they should be used as **design inspiration**, not as the sole anchor of a publication claim. [13]

What the model–brain alignment literature says about methods, conventions, and real findings

On the neural side, several things are now established. First, large-scale image-based fMRI modeling is the norm for ambitious work, and NSD is the canonical resource because it combines natural scenes, large image diversity, and enough data per subject to fit modern encoding models. Second, evaluation conventions are highly standardized: held-out prediction, reliability/noise-ceiling considerations, per-subject and per-ROI analyses, and either voxelwise encoding, RSA, or both. Third, model comparison is now a model-zoo problem rather than a two-model problem. These norms are reinforced by Brain-Score, Algonauts, and the recent wave of large comparison papers. [14]

The major findings are more nuanced than a "best architecture" story. Storrs et al. showed that once networks are appropriately trained and fitted, **diverse** DNNs predict human IT quite well, which already weakens any simplistic claim that one model family uniquely matches cortex. Zhuang et al. then showed that unsupervised contrastive models can achieve neural prediction comparable to supervised ventral-stream models, helping establish self-supervision as a serious candidate rather than a side experiment. Mehrer et al. further showed that training diet matters: an ecologically motivated dataset such as Ecoset improves higher-level cortex and perceptual judgments. This is the background against which your model choices should be made. If you benchmark only standard supervised ImageNet backbones, your project will be behind the current neural-alignment literature. [15]

Wang et al. are especially important for your project because they speak directly to high-level visual cortex and modern vision-language features. They report that CLIP-ResNet50 can explain up to $R^2 = 79\%$ of variance in held-out voxel responses in high-level visual cortex, and that comparisons controlling dataset size and diversity indicate that language feedback together with large, diverse datasets are important drivers of this improvement. This result strongly supports including CLIP-/SigLIP-like encoders in your benchmark. However, it does **not** imply that language supervision is always the key factor, because controlled comparisons later complicated that conclusion. [16]

That complication comes from Conwell et al. Their 2024 large-scale Nature Communications study is one of the best papers for steering your project because it isolates architecture, objective, and training diet at scale. Their central result is that CNNs and transformers are nearly equal on average in brain predictivity when other factors are controlled, with CNNs having only a slight edge and large overlap in score ranges. They also show that instance-level contrastive objectives are significantly more brain-predictive than non-contrastive self-supervised objectives and often comparable to matched supervised models. Perhaps most importantly, they argue that training diet has the largest and most consistent effect on brain predictivity, and that CLIP’s strongest brain results may owe more to data scale/diversity than to language alignment alone. This paper pushes strongly against any project design centered on “CNN versus ViT” as the main story. [17]

Another important correction to naive thinking comes from St-Yves et al. Their 2023 paper shows that **non-hierarchical** brain-optimized DNNs can predict V1–V4 activity as accurately as hierarchical ones, which means that good predictive accuracy does not automatically license anatomical or mechanistic claims about cortical hierarchy. In short: encoding-model success is not equivalent to proving that your model computes like the brain. This is directly relevant to your proposal. If you ultimately find that some adaptive model predicts fMRI better, the literature says you still need additional representational or interpretability analysis before claiming mechanistic equivalence. [18]

Recent work is also expanding what counts as a meaningful neural comparison. Brain Decodes Deep Nets uses factorized mappings across layer, space, scale, and channel to analyze how model representations align to cortex and reports that CLIP-like scaling preserves a more brain-like hierarchy than many alternatives. BrainSAIL localizes semantic cortical selectivity using dense ViT-like features, moving beyond global ROI scorecards toward spatially grounded semantic attribution. Adeli et al. use transformer cross-attention to build an encoder that dynamically routes retinotopic features into category-selective cortex and report state-of-the-art performance on high-level visual responses. Video work is also maturing: Sartzetaki et al. benchmark 99 models against fMRI of humans watching videos and argue that temporal modeling matters especially for early visual cortex, while task-relevant training is more important in higher-level regions. The trend is clear: the field is moving beyond simple prediction scores toward **richer mappings, region-specific analyses, and better controlled comparisons**. [19]

Finally, the representational-method literature matters for your design choices. RSA remains foundational, but newer work questions whether researchers over-interpret any single similarity measure. Recent evaluations suggest that linear CKA and Procrustes-like metrics can align better with behavioral correspondence than some classical linear-predictivity measures in certain settings, while other work calls for comparing different similarity families rather than picking one by default. This is relevant because your proposal emphasizes representational spaces. If you use only one RSA flavor and build your story on it, the result will be methodologically thinner than the current state of the field. [20]

What this literature implies for your project

The literature review above leads to a fairly clear conclusion: **your central research direction is valid, but your current pilot implementation targets the wrong abstraction level for the main claim**. The valid part is the question itself. There is enough evidence that human-like overt attention can emerge or be trained in modern vision systems, enough evidence that model–brain alignment depends on semantics/data/objective rather than architecture alone, and enough evidence that these notions of alignment can dissociate. That combination makes your core question scientifically timely. [21]

The weak part is the current empirical entry point. A benchmark that compares post-hoc saliency maps from generic classifiers against fixation maps on static image datasets is a reasonable engineering warm-up, but it is below SOTA as a scientific endpoint. The fixation literature has already moved to dedicated probabilistic models, calibration, dataset-bias analysis, and scanpath modeling. The brain-alignment literature has moved to NSD-scale encoding/RSA and large, controlled model zoos. The adaptive-attention literature is moving toward foveation, token selection, or direct gaze-conditioned training. Against that backdrop, a static Grad-CAM/gradients/rollout benchmark is best interpreted as infrastructure, not as the core experiment. [22]

So, are you on the right track? **Conceptually yes; methodologically only partially.** You are on the right track insofar as your proposal asks about the relationship between fixation-level and fMRI-level alignment, and about adaptive attention systems rather than just raw classifier accuracy. You are only partially on the right track insofar as the current pilot still treats behavioral saliency as the main stage, whereas the literature suggests it should be only one axis in a larger alignment matrix. Your own project materials already point in that direction by centering the fixation/fMRI correlation question and by recognizing that the first real NSD/Algonauts neural run is the next decisive step. filecite turn3file2 filecite turn3file3 filecite turn3file5

The most important redesign, then, is conceptual. Your project should stop asking, even implicitly, “which model is most human-like by saliency?” and instead ask: **which types of alignment co-occur, which dissociate, and does adaptive/selective computation change that pattern?** The literature strongly suggests this is the right framing. Fel et al. show that human-strategy alignment can separate from plain task accuracy; Conwell et al. show that brain predictivity is governed more by diet/objective than by architecture in isolation; St-Yves et al. show that predictive success does not force hierarchical or anatomical interpretations; and the interpretability literature shows that raw attention and explanation maps are not faithful stand-ins for model computation. Put together, these papers imply that your real contribution should be a dissociation/convergence study, not a leaderboard. [23]

Novelty, redesign, and the most promising publication-grade agenda

The good news is that your project still has a real novelty path. The field has studied **fixation prediction**, **gaze-aligned training**, **brain alignment**, and **adaptive attention** mostly in pairs. What remains relatively underdeveloped is a unified study asking whether overt human-like attention, internal routing, cortical representational alignment, and computational efficiency are the same thing or different things. That is the strongest way to redesign your project: not as another saliency paper and not as another encoding-model paper, but as a **multi-axis NeuroAI study of alignment convergence and dissociation**. [24]

In practical terms, I would redesign the project around three coupled experiments. The first should be a **behavioral alignment benchmark** that is narrower but more rigorous than the current pilot: dedicated saliency baselines such as DeepGaze IIE/MSDB, at least one scanpath model, strong controls, separated free-viewing and task-driven datasets, and multiple attribution families for CNNs and transformers, including stronger transformer-relevance methods rather than only raw attention or rollout. The second should be the **neural core**: NSD/Algonauts-based voxelwise encoding plus multiple representational metrics, with per-subject, per-ROI analyses and reliability-aware normalization. The third should be a **causal adaptive-attention intervention**: gaze-regularized token masking, FAX-style loss, NeVA-style foveation, or another controlled routing mechanism applied to a baseline ViT or CLIP-like encoder. That third step is where your project can become more than descriptive. [25]

For model selection, the literature points to a very specific ranking of priorities. First include **supervised CNNs and ViTs** as anchors. Then include **contrastive self-supervised models** and **multimodal/vision-language models** such as CLIP-like encoders, because the brain-alignment literature says these are where real differences emerge. Only after that should you emphasize biologically inspired adaptive models or hard-attention mechanisms. If you reverse that order and begin with exotic foveated models before strong SSL/VLM baselines, reviewers will reasonably ask whether you are comparing against the most informative current competitors. [26]

The strongest revised central question, in my view, is this: **Which components of human-like visual alignment—fixation behavior, internal routing, semantic selectivity, and compute allocation—predict neural alignment in visual cortex, and where are they dissociable across cortical hierarchy and task?** That formulation is strictly stronger than your current phrasing because it does not assume one monolithic “human-likeness” factor. It also matches what the literature has already shown: partial coupling, frequent dissociation, and strong roles for objective and training diet. [27]

If the immediate goal is a top AI/ML/CV submission, the most realistic paper shape is a benchmark-and-science paper with one crisp claim: either **fixation alignment and brain alignment dissociate in systematic ways**, or **a controlled adaptive-attention intervention improves one alignment axis without improving the others**. If the longer-term goal is Nature Neuroscience, Nature Machine Intelligence, or PNAS, the paper needs a stronger neuroscience claim, for example that adaptive attention selectively improves early visual cortex predictivity per FLOP, or that human-like overt attention and cortical alignment diverge sharply in higher-level semantic regions. In both cases, the novelty lies in the **cross-level relation**, not in the saliency maps themselves. [28]

The clearest “do-not-do” list from the literature is also straightforward. Do not build the paper around raw ViT attention maps as if they were human attention. Do not pool free-viewing and search datasets into a single undifferentiated score. Do not infer mechanistic brain-likeness from good encoding accuracy alone. Do not benchmark only supervised timm backbones. And do not frame the project as “CNNs versus ViTs” unless you actually control training diet and objective, because one of the best current papers has already shown that this framing is too coarse. [29]

The best edge available to your project is therefore this: **turn the current playground into a controlled cross-level benchmark, then add one causal adaptive-attention intervention.** That design is grounded in what the field has already examined, it avoids retreading the strongest existing results, and it gives you a meaningful novelty claim that can still grow into a higher-impact neuroscience story later. filecite turn3file0 filecite turn3file3 filecite turn3file15

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